



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

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Project Title: SmartFarmingAnimalsDx: AI-Embedded Predictive Farming Animals Health Monitoring and Diagnostic System Using Edge Intelligence

Introduction: Livestock farming plays a significant role in agriculture by providing milk, meat, eggs, wool, and other essential products. The health of farm animals directly influences agricultural productivity and farmers' income. However, diseases such as respiratory infections, heat stress, metabolic disorders, nutritional deficiencies, and infectious diseases often remain undetected until visible symptoms appear, leading to economic losses and poor animal welfare.

Recent advancements in Artificial Intelligence (AI), Internet of Things (IoT), wearable sensors, and Edge Computing have enabled continuous health monitoring of animals. Existing livestock monitoring systems mainly collect sensor data and rely on cloud computing for disease analysis. These systems require continuous internet connectivity, increase communication costs, and introduce delays in diagnosis.

The proposed project develops an AI-enabled wearable health monitoring system that continuously monitors physiological and behavioural parameters of farm animals using embedded sensors. Lightweight machine learning models deployed on Raspberry Pi and ESP32 edge devices perform real-time disease prediction locally without depending on cloud connectivity. Intelligent alerts are sent to farmers and veterinarians, while cloud synchronization is performed whenever internet connectivity becomes available.

Objectives:

- Design and develop a wearable smart health monitoring system for farm animals.
- Continuously monitor physiological and behavioural parameters.
- Develop machine learning models for early disease prediction.
- Deploy lightweight AI models on edge devices.
- Generate real-time health alerts for farmers and veterinarians.

- Maintain health records through local storage with optional cloud synchronization.
- Develop an affordable and portable prototype suitable for rural veterinary applications.

Software Requirements

Operating System

- Windows 11 / Ubuntu Linux

Programming Language

- Python 3.11+

Libraries & Frameworks

- TensorFlow Lite
- TensorFlow Lite Micro (TFLM)
- Scikit-learn
- NumPy
- Pandas
- OpenCV
- Flask / FastAPI
- MQTT
- SQLite

Development Tools

- Visual Studio Code
- Arduino IDE
- Raspberry Pi OS
- Google Colab
- Git & GitHub
- Jupyter Notebook

Hardware Requirements –

- Raspberry Pi 4
- ESP32 Development Board
- Heart Rate Sensor
- Body Temperature Sensor
- SpO₂ Sensor
- Respiratory Sensor
- Accelerometer & Gyroscope
- GPS Module
- Rechargeable Battery
- Battery Management Circuit
- Smart Collar
- PCB and Embedded Electronics
- Internet Connectivity (Optional for Cloud Synchronization)

Abstract: Continuous health monitoring of farm animals is essential for improving productivity, reducing disease outbreaks, and ensuring animal welfare. Traditional livestock healthcare mainly depends on manual observation and periodic veterinary examinations, often resulting in delayed diagnosis. Existing monitoring systems largely depend on cloud computing, making them unsuitable for remote farming environments with limited internet connectivity.

This project proposes an AI-enabled wearable health monitoring and diagnostic system using Edge Intelligence for real-time livestock healthcare. Embedded sensors continuously collect physiological and behavioural data including body temperature, heart rate, respiratory rate, oxygen saturation, movement patterns, and GPS location. Lightweight machine learning models deployed on Raspberry Pi and ESP32 perform on-device disease prediction without requiring continuous internet access. Intelligent alerts are immediately delivered to farmers and veterinarians whenever abnormal conditions are detected. Health records are synchronized with cloud storage whenever connectivity becomes available. The proposed system is affordable, portable, energy-efficient, and particularly suitable for precision livestock farming and rural veterinary healthcare.

Literature Survey :

Sl. No	Paper Title	Methodology	Limitation
1	IoT-Based Livestock Health Monitoring System	Uses IoT sensors for continuous health monitoring.	No AI-based disease prediction.
2	Edge AI for Smart Agriculture	Deploys lightweight AI models on edge devices.	Focuses mainly on crop monitoring.
3	Machine Learning for Livestock Disease Prediction	Uses supervised learning for disease classification.	Requires centralized cloud processing.
4	Wearable Sensors for Animal Health Monitoring	Collects physiological and behavioural parameters using wearable devices.	Performs only monitoring without intelligent diagnosis.
5	Edge Computing in Precision Livestock Farming	Implements local processing for agricultural applications.	Limited integration with wearable healthcare systems.

Existing System & its Drawback:

Existing System:Current livestock health monitoring systems primarily rely on wearable sensors and IoT devices to collect physiological data. Most systems transmit sensor data to cloud servers where machine learning models perform disease prediction. Veterinary intervention is generally initiated only after abnormalities are identified through cloud-based analysis.

Drawbacks:

- Continuous dependence on internet connectivity.
- High communication latency.
- Increased cloud processing cost.
- Limited offline functionality.
- Delayed disease diagnosis.
- Poor suitability for remote rural farms.
- High power consumption.
- Limited portability.

Proposed System & Advantages:

Proposed System: The proposed SmartFarmingAnimalsDx system integrates wearable health sensors, IoT communication, embedded edge devices, and Artificial Intelligence into a unified livestock healthcare platform. Physiological and behavioural parameters are continuously collected through wearable sensors attached to the animal. The collected data are processed locally using lightweight machine learning models deployed on Raspberry Pi and ESP32. Real-time disease prediction is performed directly on the edge device without depending on cloud connectivity. Health alerts are automatically generated for farmers and veterinarians whenever abnormal conditions are detected. Health records are stored locally and synchronized with the cloud whenever internet access becomes available.

Advantages:

- Real-time health monitoring.
- Early disease prediction.
- Offline diagnosis using Edge AI.
- Reduced latency.
- Lower communication cost.
- Energy-efficient operation.
- Affordable solution for rural livestock farming.
- Portable smart wearable system.
- Improved Animal Welfare.
- Increased Agricultural productivity.

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